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Set - A House price prediction.

step - 1 problem framing and theory Notes

Q1 ~~what~~ what is Regression? How is it different from classification? give 2 real-world indian example?

⇒ Regression is a supervised machine learning task used to predict continuous, numerical outcomes based on input features, whereas classification categorizes data into discrete, ~~pre~~ predefined classes. while a classification model might predict if an applicant will default on a loan (yes/no) a regression model predicts the exact loan amount they are eligible for.

ex :- 1) predicting house price in Ahmedabad based on factors like square footage, locality, and proximity to transit.

2) forecasting continuous retail revenue trends and inventory demand during major indian festival seasons.

Q2 what is the difference between simple Linear Regression, multiple Linear Regression and Regularised Regression (~~Ridge~~ Ridge / Lasso)

⇒ simple Linear regression models the relationship between exactly one independent variable and one continuous dependent variable by fitting a straight line.

multiple Linear regression expands on this

by utilizing two or more independent variable to predict a single continuous outcome.

Regularised Regression (Ridge / Lasso) add a mathematical penalty to the loss function to shrink coefficient values, actively preventing the model from becoming overly complex and sensitive ~~data~~ to noisy data.

Q3 Explain overfitting and underfitting. How does regularisation (L_1/L_2) help prevent ~~them~~ them?

⇒ overfitting occurs when a model memorized the training data noise and fluctuations, leading to high accuracy during training but poor generalization to unseen data.

underfitting happens when a model is too simplistic to capture the underlying patterns in the data, resulting in poor performance on both training and testing set.

Regularisation & mitigates overfitting by penalizing large model weights: L_1 (Lasso) can force less important feature weights to absolute zero while L_2 (Ridge) smoothly shrinks all weights to reduce overall model variance.

(3)

Q4 Define ~~Rm~~ RMSE, MAE and R^2 . When could you prefer RMSE over MAE?

⇒ MAE (Mean Absolute Error) :- ~~etc~~ calculates the average of the absolute differences between predicted and actual values treating all errors linearly and equally.

RMSE (Root Mean Squared Error) :- calculates the square root of the average squared error because errors are squared before averaging it heavily penalize larger deviations.

R^2 (Coefficient of Determination) :- A statistical measure (from 0 to 1) representing the ~~percentage~~ proportion of variance in the dependent variable that is predictably explained by the independent variable.

Preference : RMSE is preferred over MAE ~~is~~ in scenarios where large prediction errors are particularly costly or dangerous as it makes these outliers highly visible in the evaluation metric.

Q5 What is cross-validation (K-fold)? Why is it preferred over a single train-test split for model evaluation?

⇒ K-Fold cross-validation is a resampling procedure that divides a dataset into K equal-sized folds, training the model K-times in each iteration, a different fold is used

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for testing while the remaining folds are used for training.

it is highly preferred over a single train-test split because it guarantees that every data point is used for both training and validation across the iteration.

this comprehensive approach significantly reduces the variance associated with a random single split, providing a much more robust and trustworthy estimate of how the model will perform on truly unseen data.